LESSON 4.5

Determine the Whole Given the Part and Percent

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the whole when I know a part and the percent that part represents.

WHY THIS MATTERS Every percent problem has three pieces: the part, the whole, and the percent. So far the whole has always been handed to you. Today it is the missing piece — a sale price without the original price, a car's current value without its sticker price. Your job is to work backwards to the whole. For example, Akela paid $42 for a sweater on sale, and that was 60% of the original price. What was the original price?

REMEMBER

Determine the Whole Given the Part and Percent

Finding the Whole Given Part & Percent. Formula: Whole = Part ÷ Percent When you know a part and the percent it represents, the whole is what is missing. A double number line, a ratio table, or an equation will each get you there — the equation is the fastest once you trust it.

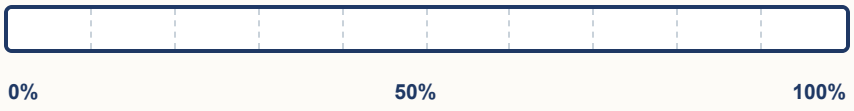
1. Identify the given part and the percentage it represents.
2. Convert the percent to a decimal or fraction.
3. Divide the part by the percent decimal to find the total whole (100%)

Words: Part *(Parte)* · Whole *(Total)* · Double number line *(Recta numérica doble)* · Equation *(Ecuación)* · Percent *(Porcentaje)*

1 **Circle the letter of the best answer. Show how you know.**

What is 10% of 320?

(A) 3.2
(B) 32
(C) 64
(D) 320



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

2 **Circle the letter of the best answer. Show how you know.**

What is 50% of 84?

(A) 34
(B) 42
(C) 50
(D) 168



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

3 **Circle the letter of the best answer. Show how you know.**

What is 20% of 150?

(A) 20
(B) 30
(C) 35
(D) 75



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

4 **Circle the letter of the best answer. Show how you know.**

Sheng's parents bought a new car last year. Its value today is $26,600, which is 70% of what they paid for it. $42 is 60% of the original price. What was the original price?

(A) $25.20
(B) $60
(C) $70
(D) $102



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

5 **Circle the letter of the best answer. Show how you know.**

Sheng's parents bought a new car last year. Its value today is $26,600, which is 70% of what they paid for it. Which equation finds the whole when 75 minutes is 15% of the school day?

(A) 0.15d = 75
(B) d = 0.15 × 75
(C) 75d = 0.15
(D) d = 75 + 15

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Sheng's parents bought a new car last year. Its value today is $26,600, which is 70% of what they paid for it. If the percent given is less than 100%, the whole must be…

(A) Smaller than the part
(B) Larger than the part
(C) Equal to the part
(D) Impossible to find

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

SENTENCE STARTER If the percent is less than 100%, the given number is a ___, so the whole must be ___ than it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each question by what it asks you to find.

Groups: Find the whole | Find the part

_____ 12 is 40% of what number?

_____ What is 40% of 12?

_____ $30 is 25% of the total. What is the total?

_____ What is 25% of $30?

2 **Write the letter of the matching item on each line.**

Match each part-and-percent pair to its whole.

_____ $42 is 60%

_____ 75 min is 15%

_____ 18 is 30%

_____ $9 is 25%

(A) $70

(B) 500 min

(C) 60

(D) $36

PART 2 Apply it

real-world problems

3 Circle the letter of the best answer. Show how you know.

$10 is 50% of what number?

(A) $5
(B) $20
(C) $60
(D) $100



0% 50% 100%

WORK

4 Circle the letter of the best answer. Show how you know.

$9 is 25% of what number?

(A) $2.25
(B) $34
(C) $36
(D) $225



0% 50% 100%

WORK

5 Circle the letter of the best answer. Show how you know.

To find the whole from a part and a percent, you…

(A) Multiply the part by the percent
(B) Add the part and the percent
(C) Subtract the percent from 100
(D) Divide the part by the percent in decimal form

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** 20 is 40% of what number?
2. **Step 1** 40% = 0.4
3. **Step 2** 20 × 0.4 = 8
4. **Step 3** The number is 8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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